EER6689 Final Project – Software Engineering

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PROGRESS REPORT I

Physics-Informed AI for 3D Dent Assessment and Repair Decision Support in Paintless Dent Repair Workshops

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Table of Contents

Executive Summary 4
1. Introduction 5
1.1 Background and Problem 5
1.2 Aim and Objectives 5
1.3 Project Scope 6
1.4 Progress Summary 6
2. Literature Survey 7
2.1 Vision-Based Damage Detection 7
2.2 Repair-Cost Estimation 7
2.3 Sri Lankan Systems and Studies 7
2.4 Measurement of Specular Surfaces and Depth Estimation 8
2.5 Panel Mechanics, Repair Practice and Uncertainty 8
2.6 Comparative Summary 8
2.7 Research Gap and Novelty 9
3. Software Requirements Specification 10
3.1 Introduction 10
3.2 Overall Description 11
3.3 External Interface Requirements 12
3.4 Functional Requirements 12
3.5 Use Cases 14
3.6 Non-Functional Requirements 16
3.7 Data Requirements 17
3.8 Requirements Traceability 17
4. System Design 18
4.1 System Overview 18
4.2 Context and Data Flow 18
4.3 Activity Diagrams 20
4.4 Sequence Diagram 21
4.5 Class Diagram 22
4.6 Database Design 23
4.7 State Machine 24
4.8 Deployment Architecture 24
4.9 Algorithm Design 25
4.10 Repair Check After PDR 27
5. Implementation Progress 28
5.1 Prototype Overview 28
5.2 3D Measurement by Deflectometry 28
5.3 Plate Solver Verification 29
5.4 PDR Suitability and Repair Method 30

5.5 Prototype API and Tests 31
5.6 Limitations of the Current Prototype 32
6. Project Plan 33
6.1 Work Breakdown Structure 33
6.2 Schedule and Milestones 33
6.3 Work Distribution 34
6.4 Risk Register 34
7. CMMI Meeting Minutes 36
7.1 Process Approach 36
7.2 Meeting Log 36
7.3 Minutes of Meetings 36
8. Conclusion and Next Steps 40
References 41

Executive Summary

This report presents the first progress of Group 03's final-year project, a physics-informed decision-support system for Paintless Dent Repair (PDR) workshops. The system measures a vehicle dent in 3D with a smartphone and a laptop screen, decides whether PDR is suitable from the strain that the repair would impose on the metal and paint, selects between push repair, dent pulling (glue-pull or stud-pull) and replacement, plans the repair, and gives the workshop technician an explained recommendation with a success probability and a cost range.

Since the proposal, the team has completed the literature survey, agreed the novelty and scope with the supervisor, written the Software Requirements Specification (SRS), produced the full set of design diagrams and built an early working prototype of the core algorithms. On simulated dents of 1–10 mm depth, including camera noise and a ±2% calibration error, the prototype reconstructs dent depth with a mean absolute error of 0.061 mm (maximum 0.138 mm, $R^2 = 0.9993$). The in-house thin-plate solver agrees with the exact solution of a clamped circular plate within 0.8–2.8% for a point load. All eight automated tests pass. Real-world validation on test panels and workshop cases is the next phase.

Table 1: Status of the items required for Progress Report I

Required item	Status	Section
Completed SRS document	Completed (IEEE 830 structure, 32 functional and 18 non-functional requirements, use cases, traceability)	3
Literature survey	Completed (international and Sri Lankan systems, measurement, mechanics, gap analysis)	2
All diagrams (overview, use case, activity, etc.)	Completed: overview, context, DFD level 1, use case, 2 activity, sequence, class, ER, state, deployment, algorithm pipeline, repair check, WBS, Gantt	4, 6
CMMI meeting minutes	Completed for six meetings; dates and attendance to be confirmed (highlighted)	7
Project plan	Completed (WBS, schedule, milestones, work distribution, risk register)	6
Implementation (encouraged)	Core algorithms, prototype REST API and 8 passing tests (simulated data)	5

1. Introduction

1.1 Background and Problem

Dents caused by door contact, minor collisions, falling objects and hail are common in Sri Lanka, and most are repaired in small independent workshops. Many can be repaired with Paintless Dent Repair (PDR), which keeps the original factory paint and is faster and cheaper than conventional repair [1]. In **push repair**, the technician reaches behind the panel with rods and pushes the metal out; when the back of the panel cannot be reached, the technician uses **dent pulling**, bonding glue tabs to the paint and pulling with a slide hammer, or welding studs where the paint is already damaged.

Whether PDR will succeed depends on the depth and curvature of the dent and on how much the metal and paint must be strained to restore the surface. None of these quantities is visible in an ordinary photograph, and none is measured in current practice: the PDR technician standard accepts a repair when no sign of the dent remains visible [2]. Two technicians can therefore reach different decisions for the same dent; an over-optimistic decision cracks the paint and causes rework, and an over-cautious one sends a repairable dent to a more expensive repair. Existing AI tools, including those used by Sri Lankan insurers, detect damage in two-dimensional photographs and estimate cost, but they do not measure the dent or assess repair feasibility [3], [4], [5], [6].

1.2 Aim and Objectives

Aim. To develop and evaluate a physics-informed decision-support system that measures vehicle dents in 3D with a smartphone and recommends a repair method, repair plan and cost range with a stated confidence level for PDR workshop technicians.

Objectives.

1. Measure dent depth and 3D shape with a smartphone using phase-measuring deflectometry, with a depth error of at most 0.5 mm against dial-gauge readings (C1, D.M.S.N. Bandara).
2. Train a neural network that outlines a dent and estimates its height map from a single photograph, as a fallback when a stripe scan is not possible (C2, S.D.K.S. David).
3. Assess PDR suitability from curvature and strain, replacing fixed threshold rules (C3, K.S. Dissanayaka).
4. Select the repair method (push repair, glue-pull or stud-pull dent pulling, or replacement), calculate force, glue-tab size and number of pulls, plan the pushes or pulls and price the plan (C4, U.S.N. Senevirathne).
5. Attach a calibrated success probability and cost interval to every recommendation (C5, K.L.N.C. Liyanage).
6. Develop a technician report application with guided capture (C5, K.L.N.C. Liyanage).
7. Validate the system on test panels and real workshop cases, including a repair check after PDR (all members).

1.3 Project Scope

The scope below was agreed with the supervisor. The system covers the dents that PDR workshops repair every day.

Table 2: Project scope

Included	Excluded
Painted metal outer body panels: doors, bonnet, roof, boot lid, wings/fenders, quarter panels Steel and aluminium panels Round dents, door dings, hail dents, creases and body-line dents Dents up to about palm size (≈150 mm; adjustable) With or without paint damage (damaged paint → a method including refinishing)	Plastic parts and bumpers Structural or frame damage Torn, punctured or rusted metal Large collision damage (the system only recommends "replace") Scratches without a dent Matte, wrapped or dull paint for deflectometry (single-photo estimate with lower confidence)

Users. The primary users are workshop technicians and PDR specialists. Workshop owners manage rates and staff. Customers and insurers receive the shared report as evidence; the system is not designed as an insurance claims tool.

1.4 Progress Summary

Table 3: Progress against the plan up to 08 November 2026

Work package	Planned output	Status
Literature survey	Review of detection, cost estimation, Sri Lankan systems, metrology, mechanics	Done
Novelty and scope	Approved focus: 3D measurement, PDR suitability, technician decision support	Done
Requirements	SRS with functional/non-functional requirements, use cases, traceability	Done
Design	Architecture, UML, DFD, ER, deployment, algorithm design	Done
C1 3D measurement	Phase shifting, temporal unwrapping, integration	Prototype on simulated data
C3 Mechanics	Curvature, strain, plate solver, suitability check	Prototype; solver verified
C4 Repair method	Method selection, force, tab size, pulls, push plan	Prototype on simulated data
C5 Confidence and API	Monte Carlo, conformal intervals, REST API	Prototype; app UI not started
C2 AI outline and depth	Synthetic renderer, U-Net	Design done; implementation next
Data collection	Test panels, dial gauge, workshop cases	Equipment being sourced

2. Literature Survey

2.1 Vision-Based Damage Detection

Early work applied convolutional neural networks to classify car damage types from photographs [7]. Later studies moved to object detection and instance segmentation. The CarDD dataset [3] provides about 4,000 high-resolution images with six damage classes, including dents, and benchmarks detection and segmentation models such as Mask R-CNN [8]. Enhanced YOLO models improve detection speed and accuracy for car damage [9], and the YOLO family is now the standard choice for real-time detection [10]. Dent-specific work [4] shows that small dents remain difficult to detect because they appear only as subtle changes in reflection and shading, and its authors identify 3D surface analysis as a necessary next step. The common limitation is that the output is a bounding box or a pixel mask: the dent is located, but its depth, shape and curvature, which decide whether PDR can be used, are not measured.

2.2 Repair-Cost Estimation

Cost-estimation studies combine detection with a pricing model. Zhang and Shin [11] and Ghita et al. [12] map detected damage to repair prices, and Ahaggach et al. [13] integrate ontology reasoning with regression models to improve prediction. Pérez-Zarate et al. [14] show that computer vision can support insurance claim handling in practice. These approaches learn the relationship between the appearance of damage and its historical price. They do not model the repair process, so they cannot explain why a dent costs what it does, and they cannot distinguish two dents that look alike in a photograph but need different repair methods.

2.3 Sri Lankan Systems and Studies

Sri Lankan insurers have digitised motor claims in several ways, summarised in the table below. These systems improve the speed of the claim process, but the assessment is performed by human engineers or inferred from 2D images, and all are designed for insurers rather than for the technician who performs the repair.

Table 4: Sri Lankan systems and studies

System / study	What it does	What it does not do
Allianz Lanka 3D motor accident assessment app (2018) [15]	Customer marks damage on a 3D car model and submits photos	No measurement; engineers assess
Asian Alliance Click2Claim (2015) [16]	Accident photos uploaded from a phone	Staff call back and assess
LOLC General Insurance V Claim [17]	Live video inspection by motor engineers	Assessment by humans only
Sri Lanka Insurance Virtual Claim Assistance (2023) [18]	Remote video-based claim assistance	No automated measurement
Fairfirst Insurance with CamCom AI [5]	AI photo inspection and instant estimate; closest system	2D only; no PDR suitability; insurer-focused

System / study	What it does	What it does not do
Jayagoda & Kasthurirathna, SLIIT, IEEE SCSE 2025 [6]	Hybrid damage and severity model; 73.3% severity accuracy	Images only; no 3D, method or cost

2.4 Measurement of Specular Surfaces and Depth Estimation

Painted panels are specular, so 3D scanning methods that rely on diffuse reflection perform poorly on them. Phase-measuring deflectometry observes how a known stripe pattern is distorted by reflection and recovers the surface slope [19]; reviews show high sensitivity to small shape changes on specular surfaces [20]. Phase-shifting algorithms [21] and Fourier fringe analysis [22] extract the phase, and gradient-field integration [23] converts slopes to a height map. Learning-based monocular depth estimation [24] and encoder–decoder networks such as U-Net [25] can predict dense depth from a single image, but they are trained on large-scale scenes, not millimetre-scale dents on reflective panels, so they need domain-specific training data and ground truth. In this project, deflectometry is the primary measurement and also supplies labels for training the single-photo model.

2.5 Panel Mechanics, Repair Practice and Uncertainty

Thin-plate theory [26] relates out-of-plane deflection to curvature, bending strain and load, and yield-line theory [27] gives the load at which a plate deforms plastically, which corresponds to the force needed to reshape a dent. These methods are standard in structural engineering but have not been applied to dent assessment from measured shapes. In PDR practice, training standards define acceptable repairs visually and do not require measurement or post-repair testing [2], and reviews highlight strong dependence on technician skill [1]. None of the reviewed systems reports calibrated uncertainty; Monte Carlo propagation [28] and conformal prediction [29] provide practical ways to attach probabilities and guaranteed-coverage intervals, and the Brier score [30] allows such probabilities to be checked.

2.6 Comparative Summary

Table 5: Comparison of the most relevant studies and solutions

Study	Approach	Key finding	Limitation
Wang et al. [3]	Detection and segmentation benchmarks	Public benchmark, 6 damage classes	2D only; no depth or feasibility
Baig et al. [4]	Deep-learning dent detection	Small dents hard to detect	No 3D; names it as future work
Ahaggach et al. [13]	Ontology + regression cost model	Reasoning improves cost prediction	Repair process not modelled
Jayagoda & Kasthurirathna [6]	Hybrid image model	73.3% severity accuracy	No 3D, method or cost
Fairfirst + CamCom [5]	Commercial AI photo inspection	Fast estimates for insurers	2D; no PDR decision

Study	Approach	Key finding	Limitation
Knauer et al. [19]	Phase-measuring deflectometry	High sensitivity on specular surfaces	Not used for dent decisions

2.7 Research Gap and Novelty

Existing research detects damage in photographs and estimates cost from appearance; optical metrology can measure specular surfaces; and plate mechanics models bending and plastic deformation. These fields have not been connected. No reviewed system measures a dent in 3D with low-cost equipment, evaluates repairability from the mechanics of the panel, chooses between pushing and pulling, or gives the technician an explained and confidence-rated decision. The novelty approved for this project is therefore:

1. **3D dent measurement from a phone** using deflectometry, with a single-photo AI fallback.
2. **PDR suitability and repair-method decision** from metal and paint strain, including push repair, glue-pull and stud-pull dent pulling, and replacement.
3. **Evidence-based decision support for workshop technicians**: measurements, reasons, confidence and a cost range for every dent.

Damage detection and cost estimation are included as supporting functions; they are not claimed as novel.

3. Software Requirements Specification

This chapter follows the structure of IEEE Std 830 [31]. Requirement priorities use the MoSCoW scale: **M** must, **S** should, **C** could.

3.1 Introduction

3.1.1 Purpose

This SRS defines the functional and non-functional requirements of the Dent Assessment System (DAS). It is intended for the project team, the supervisor and the evaluation panel, and it is the baseline for design, implementation and testing. Changes are recorded through the requirements management practice described in Section 7.

3.1.2 Product Scope

DAS is a web-based decision-support system for PDR workshops. It captures stripe-pattern images and photographs of a dented panel, reconstructs the dent in 3D, assesses PDR suitability, recommends and plans a repair method, estimates a cost range with a confidence level, and produces a shareable report. DAS supports the technician's decision; it does not perform repairs, make insurance decisions or process payments.

3.1.3 Definitions, Acronyms and Abbreviations

Table 6: Definitions and acronyms

Term	Definition
PDR	Paintless Dent Repair: restoring a dented panel without refinishing
Push repair	PDR using rods from behind the panel
Dent pulling	Pulling the dent from the outside: glue-pull (bonded tabs, slide hammer) or stud-pull (welded studs, requires refinishing)
Deflectometry	Measuring a shiny surface from the distortion of a reflected stripe pattern
Height map	Grid of surface heights z(x, y) in millimetres
Stripe set	Sequence of phase-shifted stripe images (coarse and fine, x and y)
Peak strain	Largest combined bending and membrane strain over the dent
Conformal interval	Prediction interval with a guaranteed coverage level
C1–C5	The five system components defined in Section 4.1
SUS	System Usability Scale

3.1.4 References

IEEE Std 830-1998 [31]; the project proposal and concept paper of Group 03; Sri Lanka's Personal Data Protection Act No. 9 of 2022 [32]; and the literature in Section 2.

3.1.5 Overview

Section 3.2 describes the product context, users and constraints. Sections 3.3 to 3.7 specify interfaces, functional requirements, use cases, non-functional requirements and data. Section 3.8 traces requirements to objectives, components and tests.

3.2 Overall Description

3.2.1 Product Perspective

DAS is a new, self-contained system. It runs in the browser of the technician's smartphone and of a laptop or tablet used as the stripe display, communicates over HTTPS with a cloud server, and stores data in a PostgreSQL database and object storage. The overview diagram is given in Section 4.1.

3.2.2 Product Functions

- Manage users, workshops, vehicles, panels and assessment cases.
- Pair the phone with the stripe display and capture stripe sets and photographs with quality checks.
- Reconstruct a 3D height map by deflectometry, or estimate it from a single photo.
- Compute curvature and strain and assess PDR suitability with a stated reason.
- Select the repair method and calculate force, glue-tab size, number of pulls and push or pull plan.
- Estimate cost range and success probability; flag low-confidence cases for manual inspection.
- Generate, store and share reports; record the repair outcome and re-scan after repair.

3.2.3 User Classes and Characteristics

Table 7: User classes

User class	Description	Main tasks
Technician (primary)	PDR or panel technician; uses a phone daily; limited technical training	Capture, review results, decide, share reports, re-scan
Workshop owner	Manages the workshop and pricing	Manage technicians, labour rates, materials; view history
Customer / insurer	Receives a report link; no account	View the shared report
System administrator	Project team member	Manage models, material data, system settings

3.2.4 Operating Environment

- Phone: Android 10 or later with a camera of at least 12 MP and a current Chrome browser; tripod.
- Display: laptop or tablet screen of at least 13 inches with a current browser.

- Server: Linux virtual machine with Docker; Python 3.11, FastAPI, Celery, Redis, PostgreSQL 15, S3-compatible storage.
- Network: mobile data or workshop Wi-Fi.

3.2.5 Design and Implementation Constraints

- No 3D scanner or commercial FEA software may be required.
- Inference must run on CPU in production.
- Open-source licences must be respected (e.g. AGPL-3.0 for Ultralytics YOLO).
- Personal data must be handled according to the Personal Data Protection Act [32].
- The project must be completed within the academic calendar by a team of five.

3.2.6 Assumptions and Dependencies

- Panels have a glossy painted finish suitable for deflectometry, otherwise the fallback applies.
- Published material properties and paint cracking strains are adequate as starting values and will be calibrated on test panels.
- Partner workshops agree to provide cases and technician opinions.
- The technician can place a tripod and screen near the panel.

3.3 External Interface Requirements

Table 8: External interfaces

Interface	Requirement
User interface	Mobile-first web pages: case list, new case form, capture screen with live guidance, result screen (3D view, verdict, method, plan, cost range, confidence), report page. Large touch targets; English with Sinhala labels planned.
Stripe display	Full-screen page showing coarse and fine phase-shifted stripes in x and y, controlled by the phone through the server; brightness and pattern pitch configurable.
Hardware	Phone camera through the browser camera API; laptop/tablet screen; tripod; printed reference card; digital dial indicator and force gauge for validation (not connected to the system).
Software	REST/JSON API (FastAPI) documented with OpenAPI; PostgreSQL; S3-compatible object storage; Redis job queue; PyTorch, OpenCV, NumPy/SciPy in the analysis workers.
Communication	HTTPS (TLS 1.2+) for all traffic; WebSocket or polling for pairing and job status; time-limited signed links for shared reports.

3.4 Functional Requirements

Table 9: Functional requirements

ID	Requirement	Priority	Comp.
FR-01	Users shall register and log in with email and password; roles: technician, owner, admin.	M	C5

ID	Requirement	Priority	Comp.
FR-02	The owner shall manage technician accounts, labour rate and material prices.	S	C5
FR-03	The technician shall create a case with vehicle, panel type, material, thickness, rear access and paint state.	M	C5
FR-04	The system shall pair the phone with the stripe display using a short code or QR code.	M	C5
FR-05	The system shall display coarse and fine phase-shifted stripe sequences in x and y and trigger the phone capture for each pattern.	M	C1/C5
FR-06	The system shall reject a stripe set with blur, movement, over-exposure or low fringe modulation and ask for a retake.	M	C1
FR-07	The technician shall capture one ordinary photo with a reference card for scale and detection.	M	C5
FR-08	The system shall detect and localise the dent in the photo (supporting function).	S	C2
FR-09	The system shall compute the wrapped phase with the N-step phase-shifting algorithm.	M	C1
FR-10	The system shall unwrap the fine phase using the coarse phase (temporal unwrapping).	M	C1
FR-11	The system shall convert phase to slope and integrate slopes into a height map in millimetres.	M	C1
FR-12	The system shall report dent depth, equivalent diameter and outline.	M	C1
FR-13	When a stripe scan is not possible, the system shall estimate the dent mask and height map from the single photo.	S	C2
FR-14	The system shall mark single-photo results with a lower confidence level.	M	C2/C5
FR-15	The system shall compute principal curvatures, bending strain and membrane strain from the height map.	M	C3
FR-16	The system shall detect creases from the curvature distribution.	S	C3
FR-17	The system shall decide PDR suitability by comparing peak strain with the paint cracking limit and state the reason.	M	C3
FR-18	The system shall solve the thin-plate equation for unit pushes to obtain the panel influence matrix.	S	C3
FR-19	The system shall recommend push repair, glue-pull, stud-pull with refinishing, or replacement, with reasons.	M	C4
FR-20	The system shall calculate the yield-line pull force for the panel material and thickness.	M	C4
FR-21	For glue-pull, the system shall calculate the minimum glue-tab area and reject glue-pull if the tab does not fit the dent.	M	C4
FR-22	For dent pulling, the system shall estimate the number of slide-hammer pulls.	M	C4
FR-23	For push repair, the system shall plan push points and order them.	S	C4
FR-24	The system shall price the plan from set-up time, time per push or pull, labour rate and materials.	M	C4

ID	Requirement	Priority	Comp.
FR-25	The system shall compute a success probability by Monte Carlo propagation of input uncertainties.	M	C5
FR-26	The system shall give a cost interval with stated coverage using conformal prediction.	M	C5
FR-27	The system shall flag low-confidence results for manual inspection.	M	C5
FR-28	The system shall generate a report with 3D view, measurements, verdict, reasons, method, plan, probability and cost range.	M	C5
FR-29	The technician shall share a report through a time-limited link or PDF.	S	C5
FR-30	The technician shall re-scan after repair and the system shall report residual depth.	S	C1/C5
FR-31	The technician shall record the repair outcome (method used, pulls, time, cost).	M	C5
FR-32	The admin shall upload model versions and edit material properties.	C	C5

3.5 Use Cases

The use case diagram below shows the actors and use cases; the main use cases are then specified in detail.

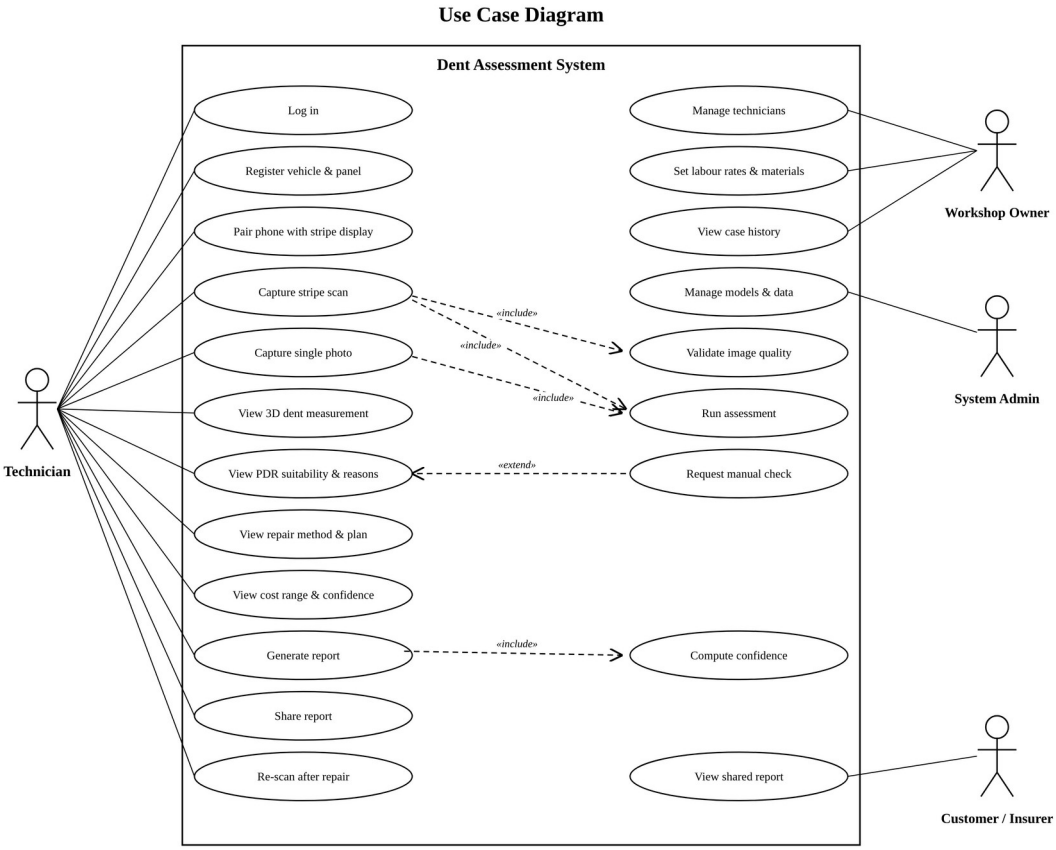


Figure 1: Use case diagram

Table 10: Use case UC-01: Capture stripe scan

Actor	Technician
Precondition	Logged in; case created; phone paired with the stripe display; tripod placed
Main flow	1. Technician starts capture. 2. Display shows each stripe pattern. 3. Phone captures each image automatically. 4. System checks blur, movement, exposure and fringe modulation. 5. System confirms the set.
Alternative flow	4a. Quality check fails → system explains the cause and asks for a retake. 4b. Low modulation (matte/wrapped paint) → system proposes single-photo capture (UC-02 fallback).
Postcondition	Valid stripe set stored for the case
Requirements	FR-04, FR-05, FR-06

Table 11: Use case UC-02: Run assessment

Actor	Technician (initiates); analysis server
Precondition	Valid stripe set or single photo stored
Main flow	1. Technician submits. 2. Server queues a job. 3. C1 (or C2) produces a height map. 4. C3 computes strain and suitability. 5. C4 selects the method and plan. 6. C5 adds probability and cost range. 7. Report is stored and shown.
Alternative flow	3a. Height map invalid → job fails with reason; case returns to capture. 6a. Probability between 0.2 and 0.8 → manual check flag.
Postcondition	Recommendation and report stored
Requirements	FR-09 to FR-28

Table 12: Use case UC-03: View repair recommendation

Actor	Technician
Precondition	Assessment completed
Main flow	1. Technician opens the result. 2. System shows 3D view, depth, size and peak strain. 3. System shows verdict and reasons. 4. System shows method, force, tab size, pulls or push plan. 5. System shows success probability and cost range.
Alternative flow	2a. Single-photo source → lower-confidence banner. 5a. Manual-check flag → system recommends inspection before quoting.
Postcondition	Technician decides the repair
Requirements	FR-12, FR-17, FR-19 to FR-28

Table 13: Use case UC-04: Share report

Actor	Technician; customer or insurer (receiver)
Precondition	Report generated
Main flow	1. Technician chooses share. 2. System creates a signed link valid for a set time, or a PDF. 3. Receiver opens the report without an account.
Alternative flow	3a. Link expired → receiver sees an expiry message.
Postcondition	Share recorded

Requirements	FR-29, NFR-09

Table 14: Use case UC-05: Re-scan after repair

Actor	Technician
Precondition	Repair carried out; case in “Repaired” state
Main flow	1. Technician captures a new stripe set. 2. System reconstructs the height map. 3. System reports residual depth and compares it with the original. 4. Technician records the outcome.
Alternative flow	3a. Residual depth above target → system suggests further work.
Postcondition	Outcome stored; case closed
Requirements	FR-30, FR-31

3.6 Non-Functional Requirements

Table 15: Non-functional requirements

ID	Category	Requirement and measure
NFR-01	Accuracy	Dent depth error ≤ 0.5 mm against dial-gauge profiles on test panels
NFR-02	Accuracy	Strain-based suitability agrees with technicians better than fixed rules (Cohen’s kappa [33])
NFR-03	Calibration	90% cost intervals cover ≥ 90% of actual costs; Brier score reported
NFR-04	Performance	Capture to report in ≤ 60 s on a 4-vCPU server; stripe capture ≤ 20 s
NFR-05	Performance	API responses (non-analysis) ≤ 500 ms at the 95th percentile
NFR-06	Usability	SUS score ≥ 68 with at least eight technicians [34]; assessment completed without help
NFR-07	Usability	Every verdict shows a plain-language reason
NFR-08	Reliability	Failed jobs are reported with a cause; no silent failure; low confidence flagged
NFR-09	Security	HTTPS only; JWT authentication; role-based access; shared links expire
NFR-10	Security	Passwords hashed (bcrypt/argon2); secrets outside source code
NFR-11	Privacy	Number plates and faces blurred before storage; plate stored only as a hash; consent recorded [32]
NFR-12	Availability	≥ 99% during the evaluation period
NFR-13	Maintainability	Components behind typed interfaces; unit test coverage ≥ 70% for analysis modules
NFR-14	Portability	Runs in Chrome on Android 10+; server deployable with Docker Compose
NFR-15	Scalability	Analysis workers scale horizontally through the job queue
NFR-16	Fairness	Accuracy reported separately for paint colours, lighting and panel types
NFR-17	Auditability	Each result stores model version, parameters and inputs
NFR-18	Legal	Open-source licences respected; no proprietary insurer data used

3.7 Data Requirements

The logical data model is shown in the ER diagram (Section 4.6). Images, stripe sets and height maps are stored in object storage and referenced by key; structured results are stored in PostgreSQL. Each measurement records its source (deflectometry or single photo), grid spacing and quality flags. Each recommendation stores method, force, tab size, pulls, plan, reasons, success probability and cost interval. Repair outcomes are stored to evaluate and improve the models. Personal data is minimised: only a hash of the number plate is kept, and data is retained only for the project and research analysis.

3.8 Requirements Traceability

Table 16: Traceability: objectives, requirements, components and tests

Objective	Requirements	Component / owner	Verification
O1 3D measurement	FR-05, 06, 09–12, 30; NFR-01, 04	C1 · Bandara	Dial-gauge comparison; unit tests T1, T2
O2 Single-photo depth	FR-08, 13, 14; NFR-16	C2 · David	IoU, height-map MAE on real photos
O3 PDR suitability	FR-15–18; NFR-02, 07	C3 · Dissanayaka	Solver verification T3; agreement with technicians T4
O4 Method and plan	FR-19–24	C4 · Senevirathne	Rule tests T5, T6; actual repairs
O5 Confidence	FR-25–27; NFR-03	C5 · Liyanage	Coverage test T7; Brier score
O6 Technician app	FR-01–04, 07, 28, 29, 31; NFR-05, 06, 09–11	C5 · Liyanage	API test T8; SUS study
O7 Validation	FR-30, 31; NFR-01–03, 06	All members	Test panels, workshop cases, repair check

4. System Design

4.1 System Overview

The figure below gives the overview of the entire system. The technician's phone and a laptop screen capture the dent; the analysis server reconstructs it in 3D (C1, or C2 as fallback), checks PDR suitability (C3), selects and plans the repair (C4), and attaches confidence (C5). The report goes to the technician, who makes the final decision and may share it with the customer or insurer.

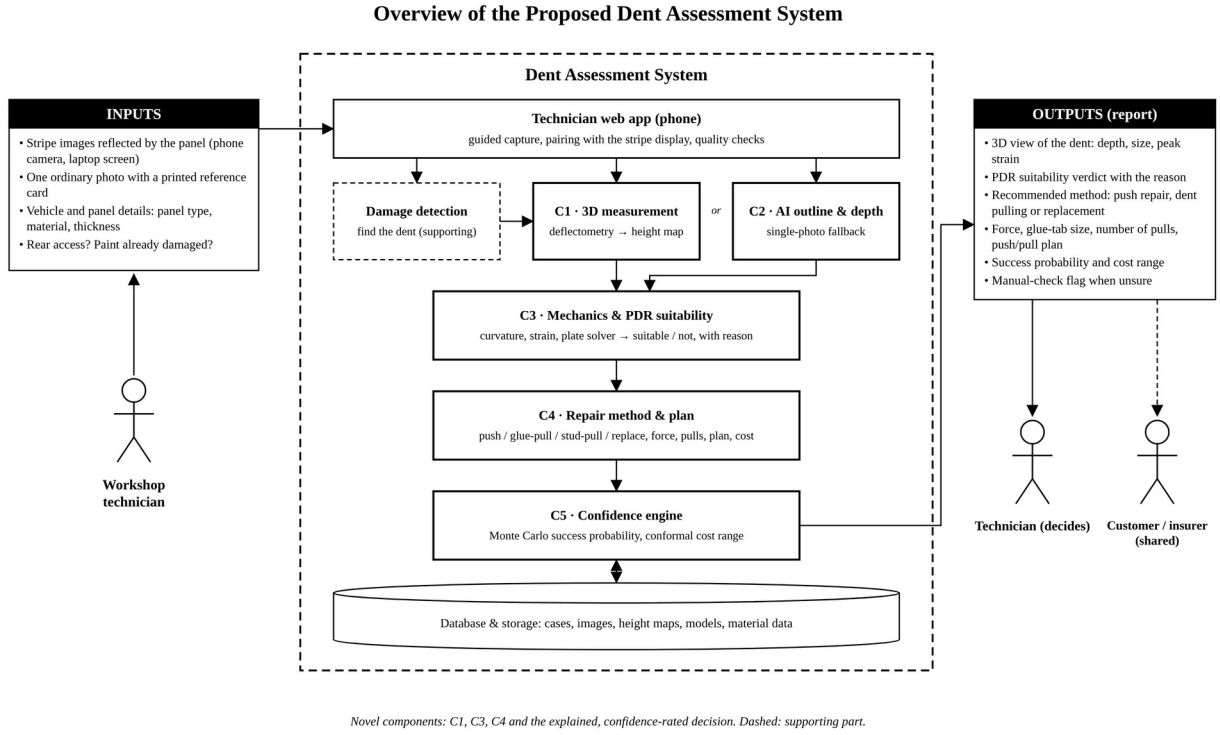


Figure 2: Overview diagram of the proposed system

4.2 Context and Data Flow

The context diagram shows the four external entities, and the level-1 DFD decomposes the system into eight processes and five data stores; note that the single-photo route (process 3.0) and the deflectometry route (2.0) both deliver a height map to the strain analysis (4.0).

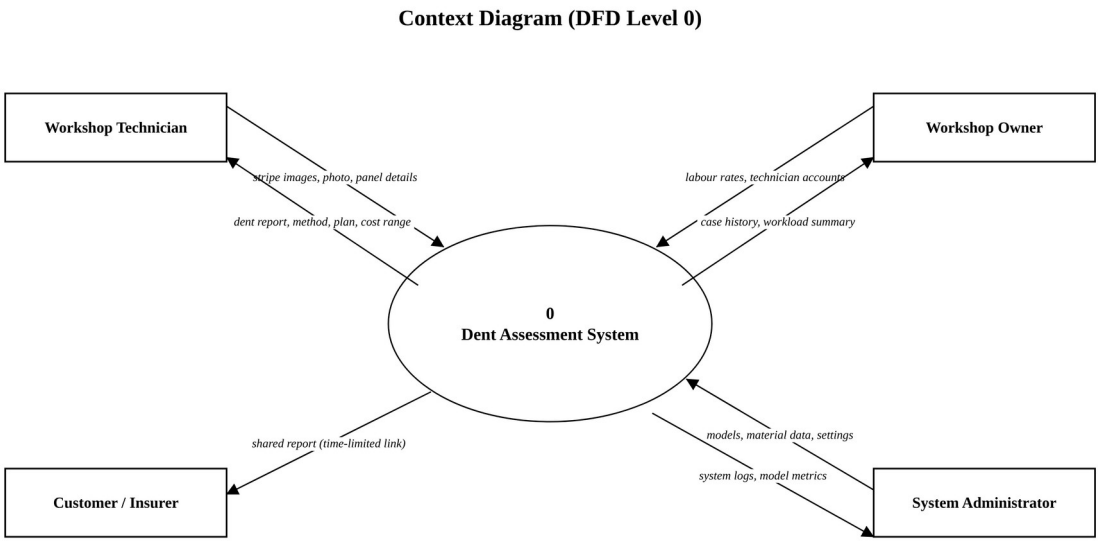


Figure 3: Context diagram (DFD level 0)

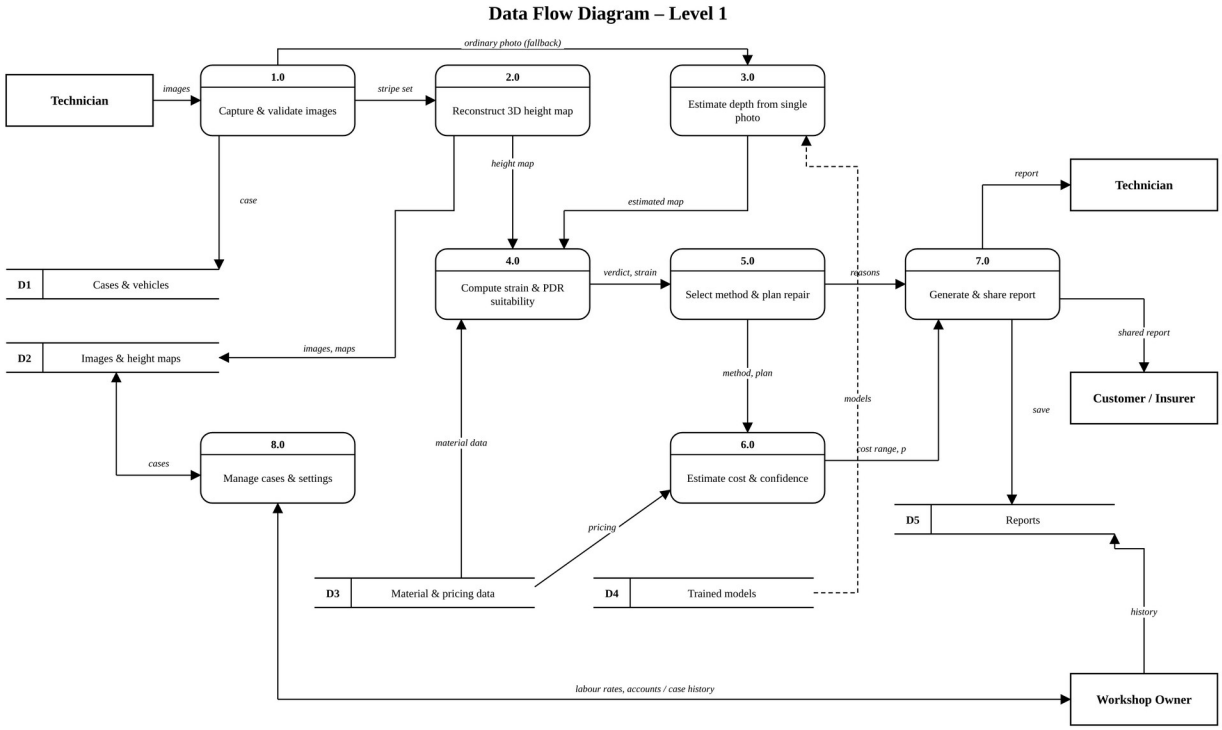


Figure 4: Data flow diagram, level 1

4.3 Activity Diagrams

The first activity diagram shows the end-to-end assessment workflow across the technician, the app and the server, including the image quality loop and the manual-check decision. The second details the repair-method selection logic of Component 4.

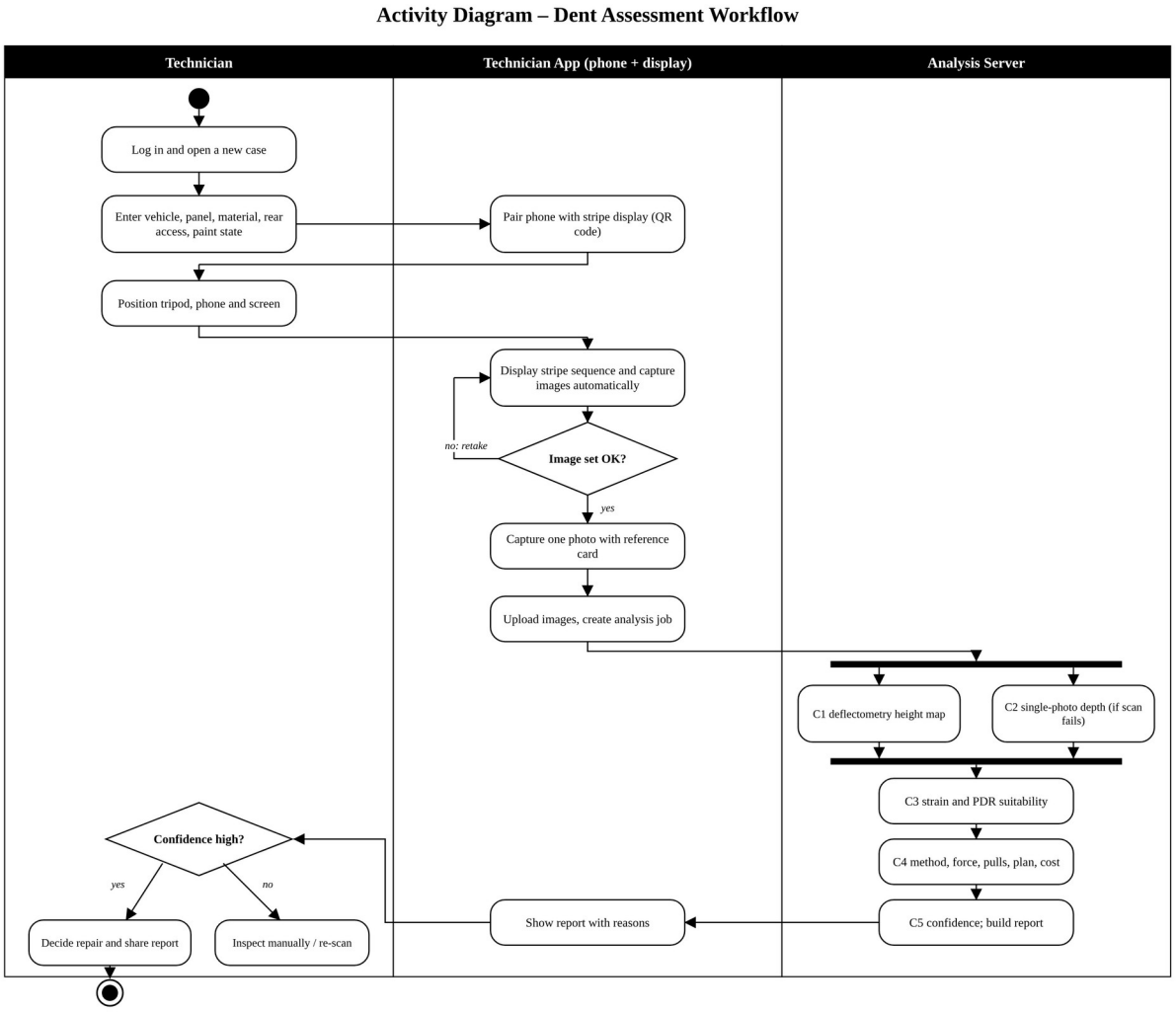


Figure 5: Activity diagram: dent assessment workflow

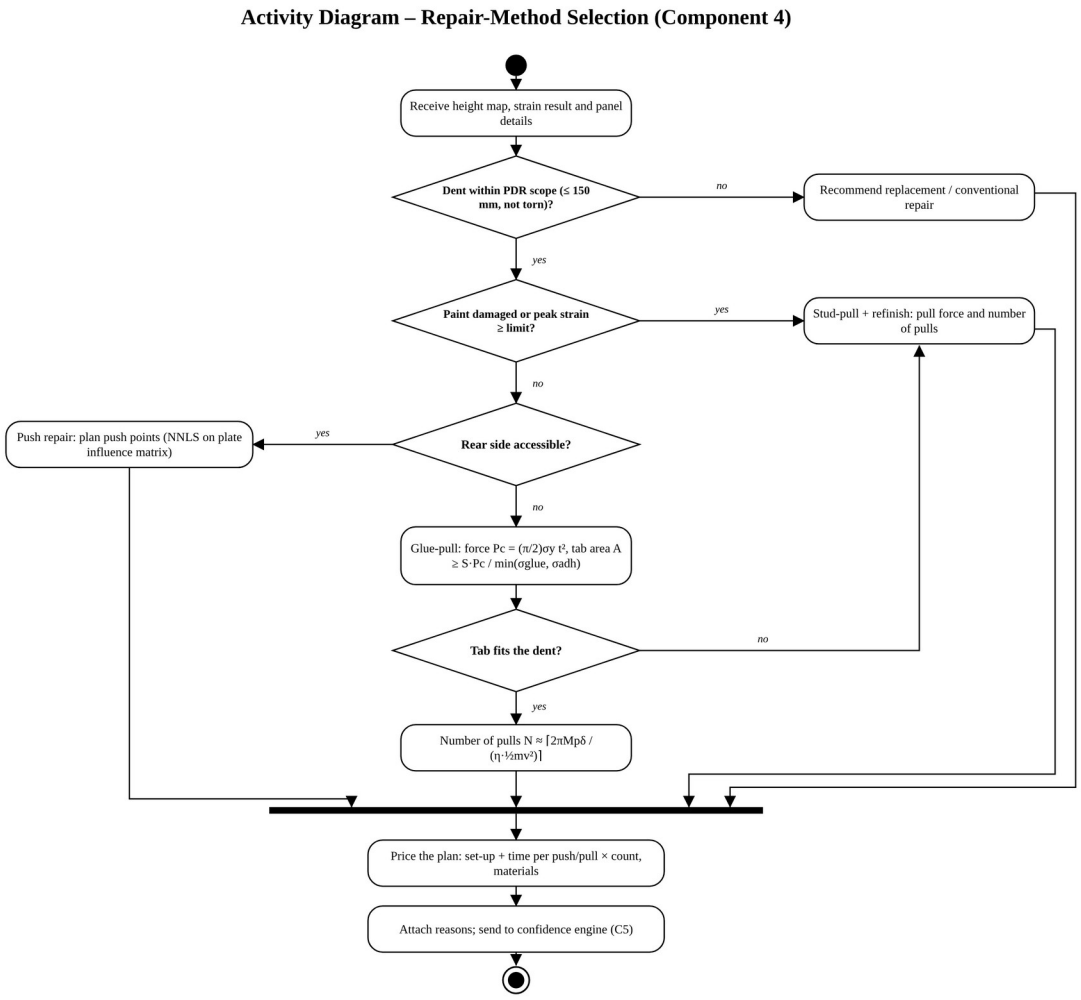


Figure 6: Activity diagram: repair-method selection

4.4 Sequence Diagram

The sequence diagram shows the interactions for one assessment. Analysis runs asynchronously: the API returns a job identifier immediately (HTTP 202) and the app polls until the report is ready, so long computations do not block the user interface.

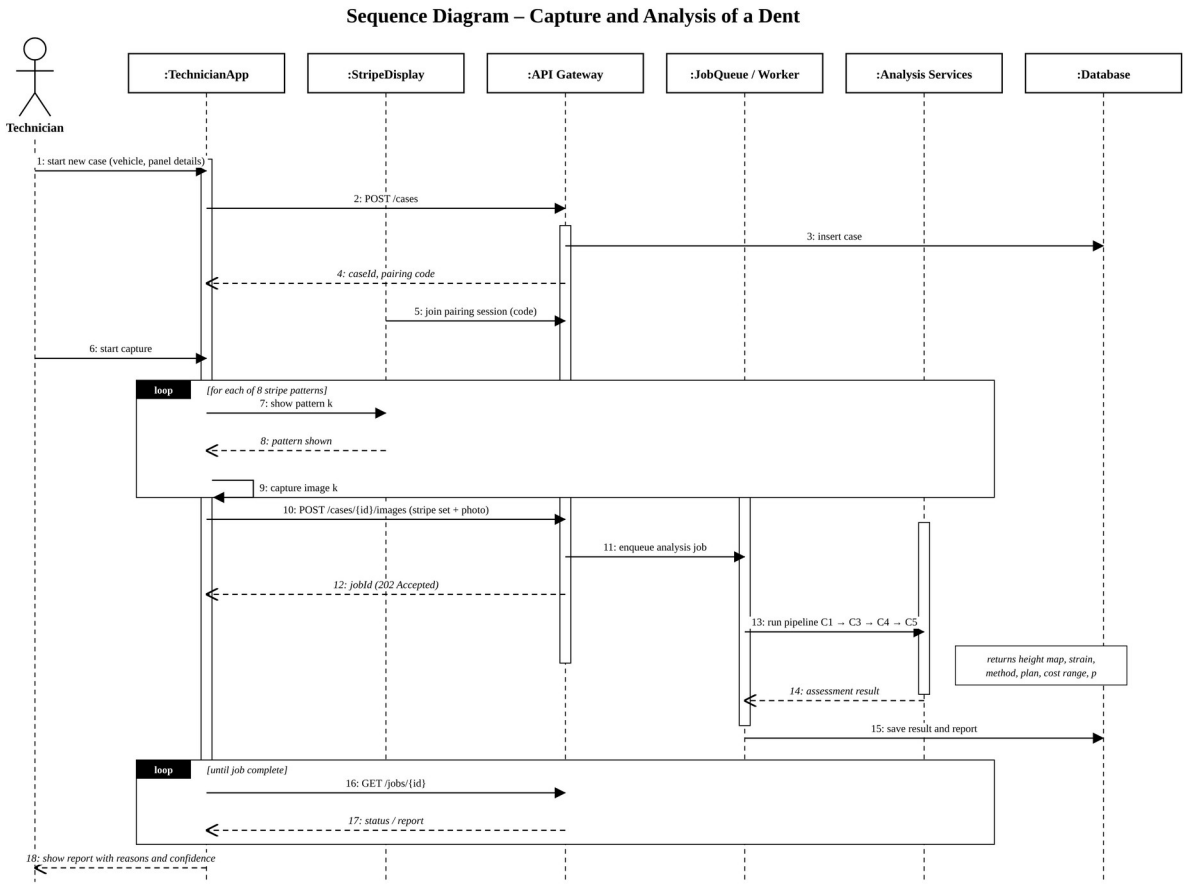


Figure 7: Sequence diagram: capture and analysis of a dent

4.5 Class Diagram

The class diagram shows the domain and analysis classes. The HeightSource interface is implemented by Deflectometry and SinglePhotoUNet, so the mechanics and planning code is independent of how the height map was obtained.

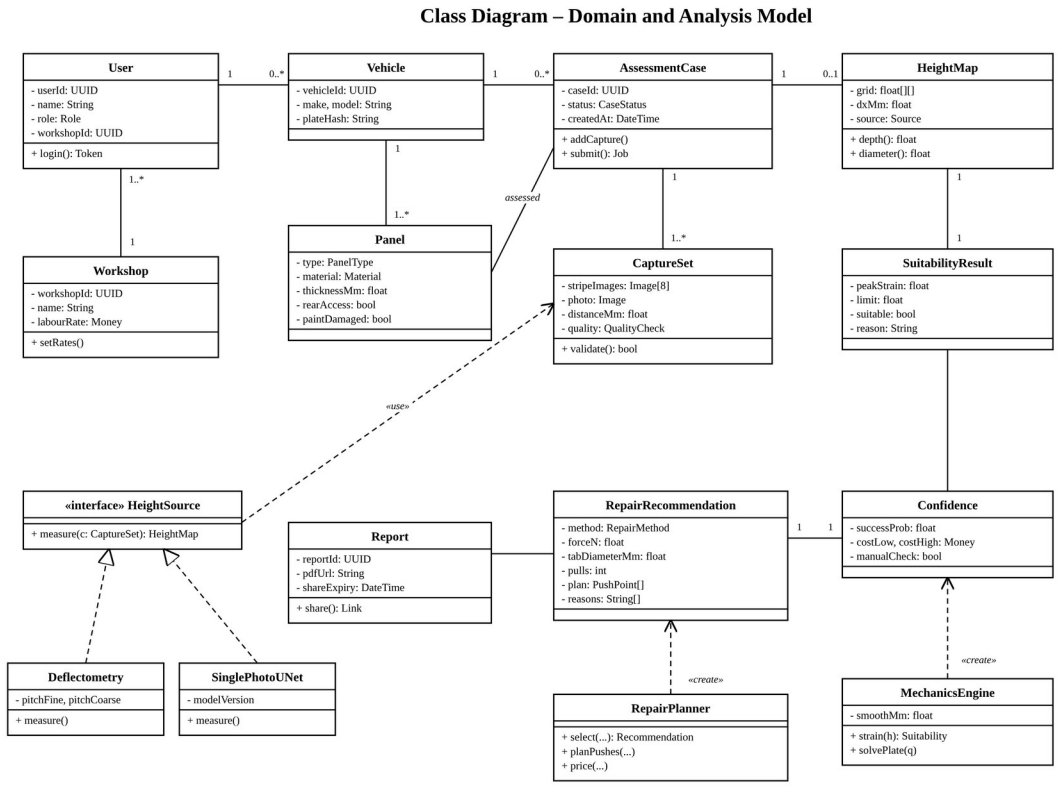


Figure 8: Class diagram

4.6 Database Design

The ER diagram shows the database design. The MEASUREMENT table separates the measured shape from the case so that a case can have both an initial scan and a re-scan after repair; REPAIR_OUTCOME stores what was actually done, which is needed to evaluate and calibrate the system.

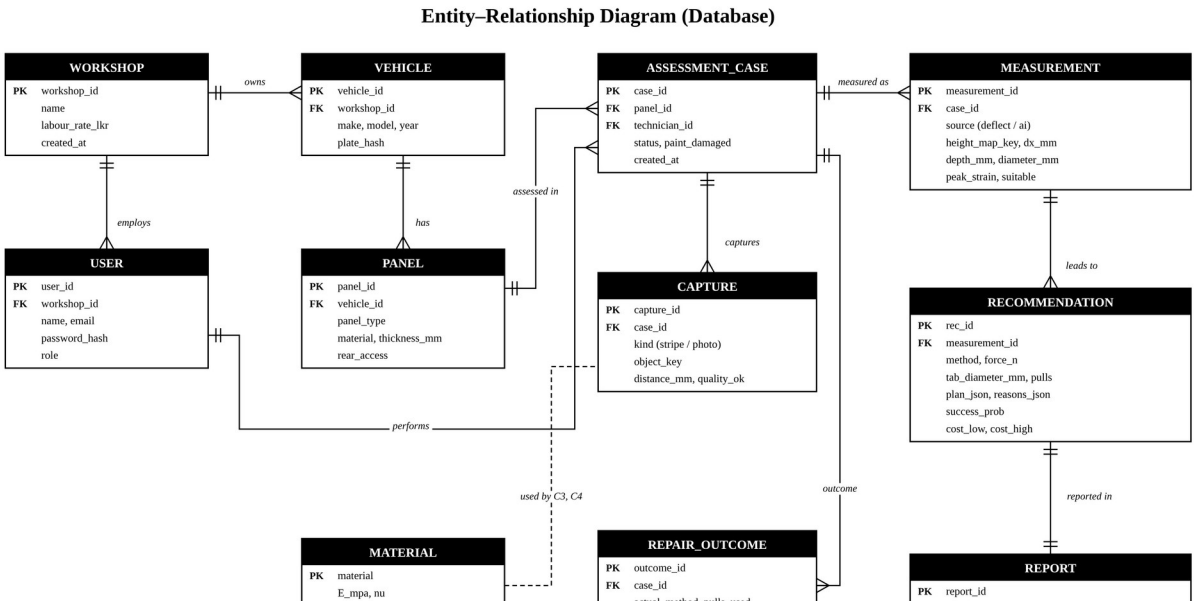


Figure 9: Entity–relationship diagram

4.7 State Machine

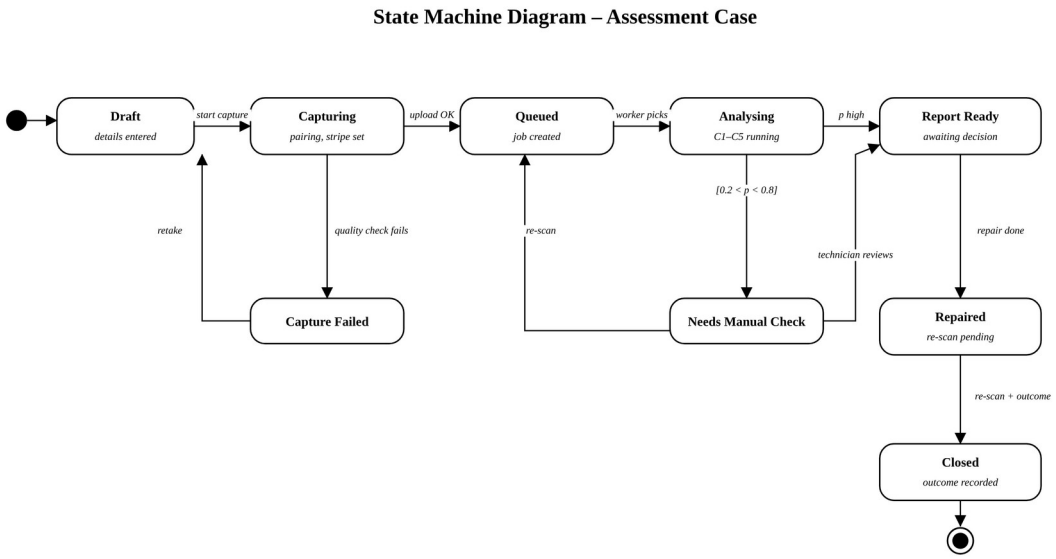


Figure 10: State machine diagram of an assessment case

4.8 Deployment Architecture

The server runs as Docker containers on one cloud virtual machine: an Nginx proxy for HTTPS and static files, the FastAPI gateway, Celery workers that run the analysis components, Redis as the job queue, PostgreSQL and MinIO object storage. Clients are browsers only, so no app-store installation is needed.

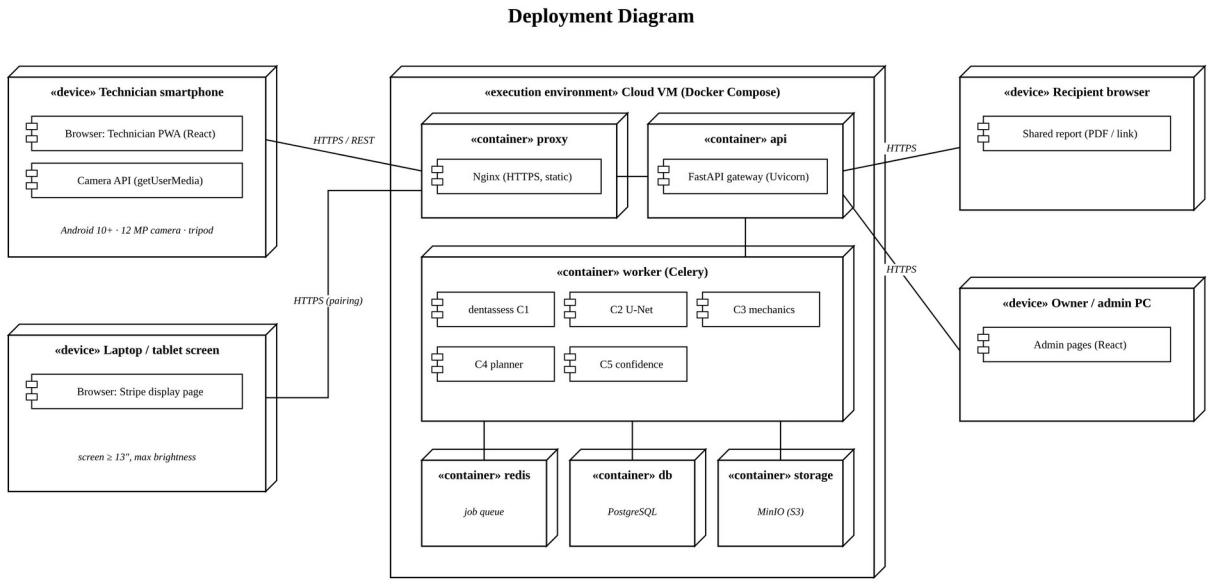


Figure 11: Deployment diagram

4.9 Algorithm Design

Component 1: 3D measurement

The pipeline is shown below. The phase at each pixel follows from the four phase-shifted intensities [21]; the fine phase is unwrapped using the coarse phase; the phase change gives the slope, where p is the stripe period and d the screen-to-panel distance; and the height map is obtained by solving a Poisson equation with the FFT [23].

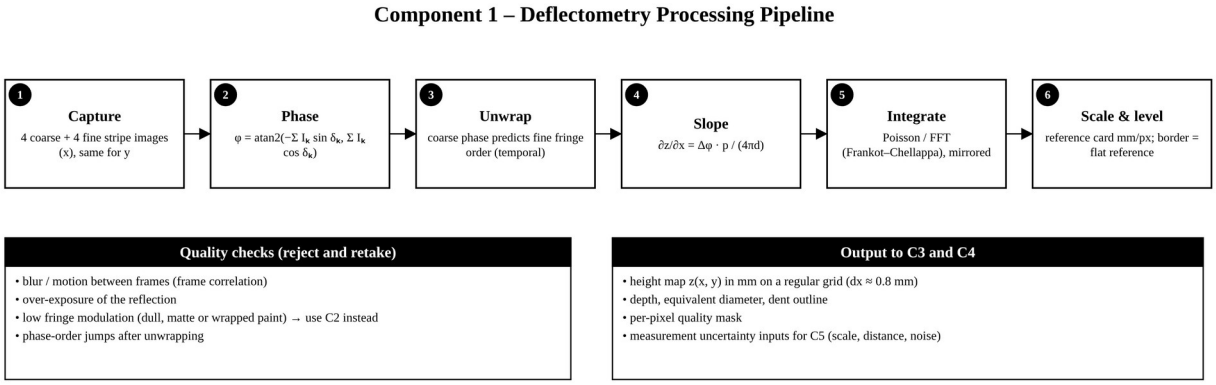


Figure 12: Component 1 deflectometry processing pipeline

$$\phi(u, v) = \operatorname{atan2}(I_3 - I_1,\ I_0 - I_2)$$

$$\frac{\partial z}{\partial x} \approx \frac{p\,\Delta\phi_x}{4\pi d}, \qquad \frac{\partial z}{\partial y} \approx \frac{p\,\Delta\phi_y}{4\pi d}$$

$$\nabla^2 z = \frac{\partial p_s}{\partial x} + \frac{\partial q_s}{\partial y}$$

Component 2: single-photo depth

A U-Net [25] with two heads predicts a dent mask and a height map, trained on synthetic dents and fine-tuned on real photos labelled by deflectometry, with the combined loss:

$$\mathcal{L} = \mathcal{L}_{\text{Dice}}(M, \hat{M}) + \lambda_1 \|z - \hat{z}\|_1 + \lambda_2 \|\nabla z - \nabla \hat{z}\|_1$$

Component 3: strain and PDR suitability

After light smoothing, bending strain is estimated from the maximum principal curvature and membrane strain from the slope; PDR is suitable when the total stays below the paint cracking strain with a safety margin. The thin-plate equation is solved by finite differences with a 13-point stencil and clamped edges [26].

$$\varepsilon_b = \frac{t}{2}\kappa_{\max}, \qquad \varepsilon_m \approx \frac{1}{2}|\nabla z|^2, \qquad \varepsilon = \varepsilon_m + \varepsilon_b < \varepsilon_{\text{crack}}$$

$$D\,\nabla^4 w = q, \qquad D = \frac{E\,t^3}{12\,(1-\nu^2)}$$

Component 4: repair method, force and pulls

The collapse force follows from yield-line theory [27]; the glue tab must keep the pull stress below the glue strength and the paint adhesion; and the number of slide-hammer pulls follows from the plastic work needed to restore a dent of depth δ. Push points are planned with non-negative least squares on the plate influence matrix.

$$M_p = \frac{\sigma_y\, t^2}{4}, \qquad P_c = 2\pi M_p = \frac{\pi}{2}\sigma_y\, t^2$$

$$A_{\text{tab}} \geq \frac{S\,P_c}{\min(\sigma_{\text{glue}},\ \sigma_{\text{adh}})}$$

$$N_{\text{pulls}} \approx \left\lceil \frac{2\pi M_p\,\delta}{\eta \cdot \frac{1}{2} m v^2} \right\rceil$$

Component 5: confidence

The success probability is the share of Monte Carlo runs [28] in which the peak strain stays below the sampled cracking strain, and the cost interval uses split conformal prediction [29]:

$$p_{\text{success}} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\left[\varepsilon_{\max}^{(i)} < \varepsilon_{\text{crack}}^{(i)}\right]$$

$$\hat{q} = \text{Quantile}_{\lceil (n+1)(1-\alpha) \rceil / n}\left\{ |y_i - \hat{y}_i| \right\}_{i=1}^{n}$$

4.10 Repair Check After PDR

Workshops do not test strength after PDR [2]. The system therefore uses a non-destructive field check on customers' vehicles and proves strength with laboratory tests on scrap test panels only.

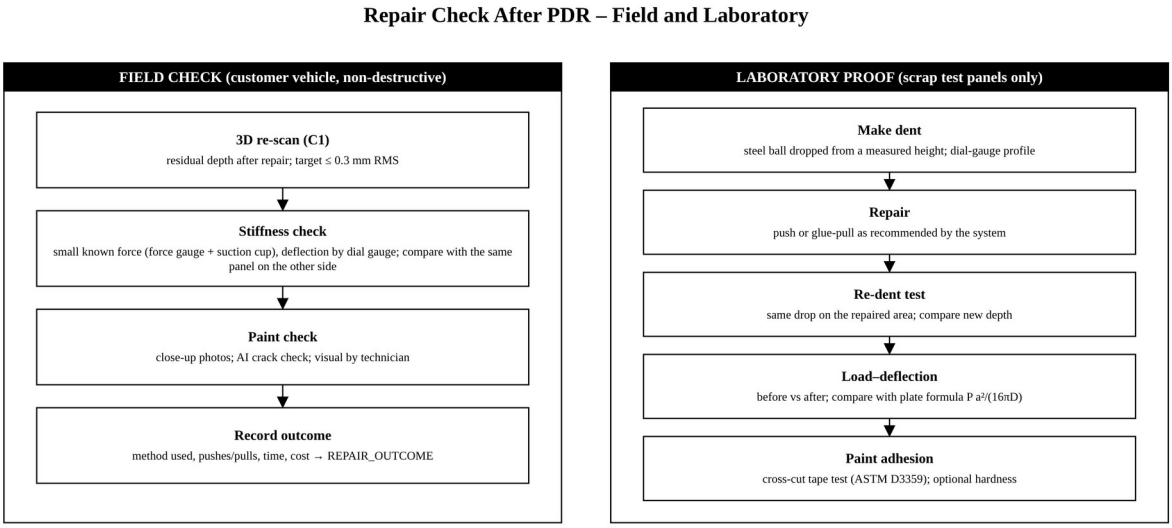


Figure 13: Repair check after PDR: field and laboratory

5. Implementation Progress

5.1 Prototype Overview

An early prototype of the core algorithms has been implemented in Python as the package *dentassess*, with a FastAPI service and automated tests. All results in this chapter use **simulated dents with exact height maps**; they verify the algorithms and the code, not yet the real-world accuracy, which will be measured on test panels in the next phase.

Table 17: Implemented modules

Module	Implemented functions	Status
synthetic.py	Synthetic dents (depth, size, sharpness, elongation) and panel waviness	Done
deflectometry.py	Stripe simulation, N-step phase, two-period temporal unwrapping, slope, Frankot–Chellappa integration, levelling	Done (simulated)
mechanics.py	Principal curvatures, bending and membrane strain, crease score, suitability check, clamped-plate FD solver, verification	Done
repair.py	Yield-line force, glue-tab area, number of pulls, method selection with reasons, NNLS push planning	Done (rules to be calibrated)
confidence.py	Monte Carlo success probability, split-conformal half-width, Brier score	Done
pipeline.py	End-to-end assessment returning an explained report	Done
app/main.py	FastAPI: /api/health, /api/assess/synthetic, /api/solver/verify	Prototype
tests/test_core.py	8 automated tests (pytest)	All passing

5.2 3D Measurement by Deflectometry

The deflectometry module was tested on ten simulated dents from 1 to 10 mm deep. Each measurement used a fine stripe period of 40 mm and a coarse period of 400 mm at a 150 mm screen distance, camera noise of 5% of the fringe amplitude, and a random calibration error of up to ±2% in the assumed screen distance. The figures below show the processing stages and the accuracy.

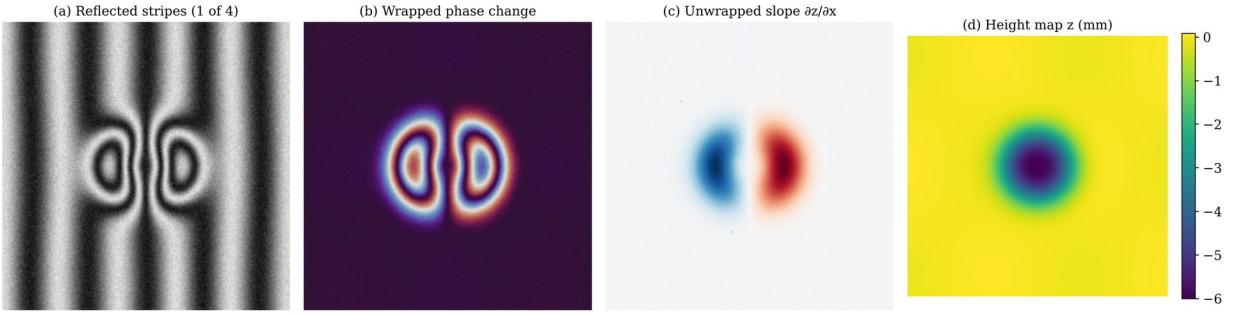


Figure 14: Prototype deflectometry stages on a simulated 6 mm dent

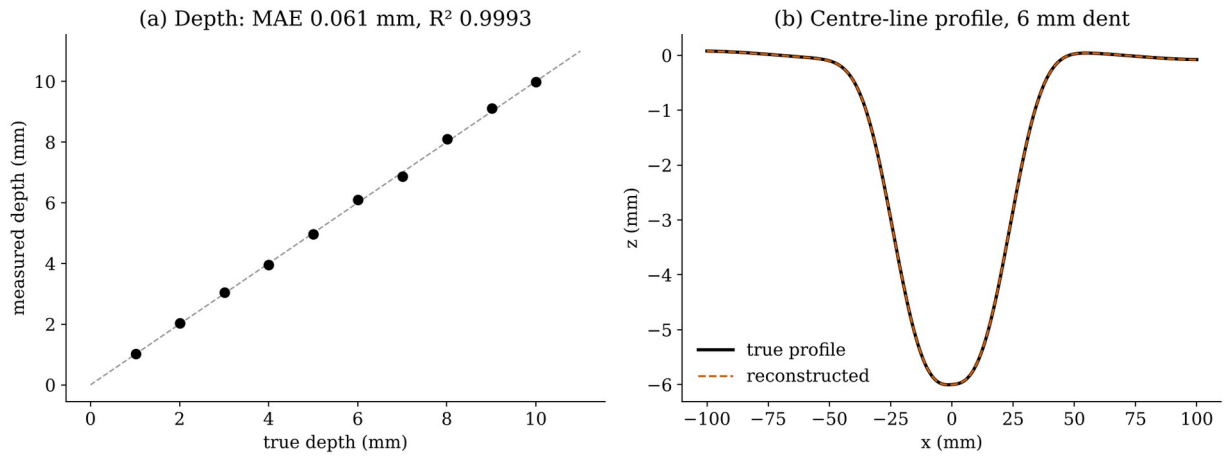


Figure 15: Measured versus true depth (10 simulated dents) and a centre-line profile

Table 18: Depth reconstruction results (simulated, noise 5%, distance error up to ±2%)

True depth (mm)	Measured (mm)	Error (mm)	Map MAE (mm)	Distance error
1.014	1.020	0.006	0.002	+0.50%
2.008	2.039	0.031	0.003	+1.59%
3.014	3.050	0.035	0.006	+1.10%
4.003	3.958	0.045	0.003	−1.10%
5.006	4.965	0.041	0.004	−0.80%
6.005	6.095	0.090	0.006	+1.49%
7.004	6.866	0.138	0.017	−1.98%
8.002	8.105	0.103	0.005	+1.28%
9.005	9.110	0.105	0.009	+1.19%
10.003	9.992	0.011	0.002	−0.13%
Mean		0.061		$R^2 = 0.9993$

The errors are dominated by the calibration error, which scales the slope and therefore the depth, rather than by camera noise. This confirms that accurate measurement of the screen distance and of the reference-card scale is the key calibration task for the real system.

5.3 Plate Solver Verification

The finite-difference solver was verified against the exact centre deflection of a clamped circular plate (radius 50 mm, 0.7 mm steel). The error decreases as the grid is refined, as expected for a consistent scheme.

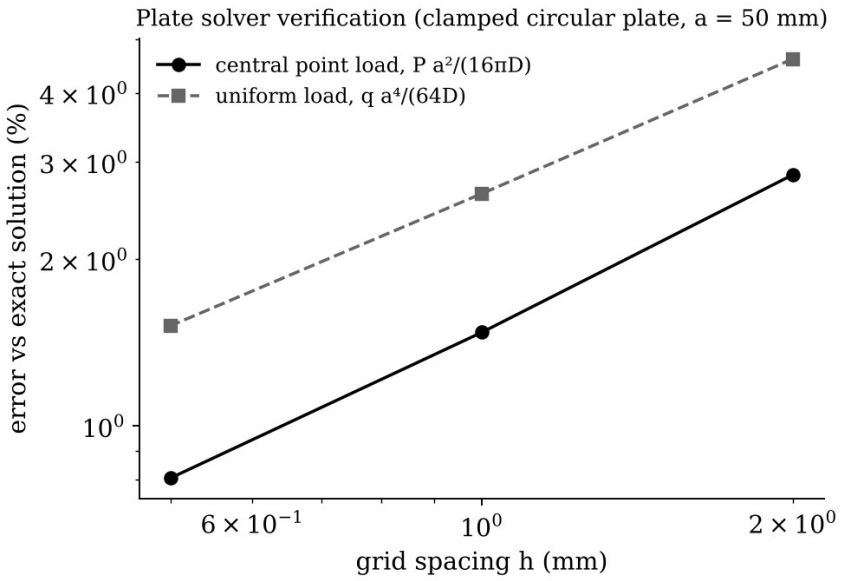


Figure 16: Plate solver error versus grid spacing

Table 19: Plate solver verification

Grid spacing h	Point load: error vs P a²/(16πD)	Uniform load: error vs q a⁴/(64D)
2.0 mm	2.84%	4.60%
1.0 mm	1.47%	2.63%
0.5 mm	0.80%	1.52%

5.4 PDR Suitability and Repair Method

The figure below shows how the peak strain varies with dent depth and size for 0.7 mm steel. Shallow, wide dents fall below the strain limit (suitable), while deep, narrow dents exceed it. The 0.8% limit used here (80% of an assumed 1% cracking strain) is a placeholder; the real limit will be calibrated on painted test panels.

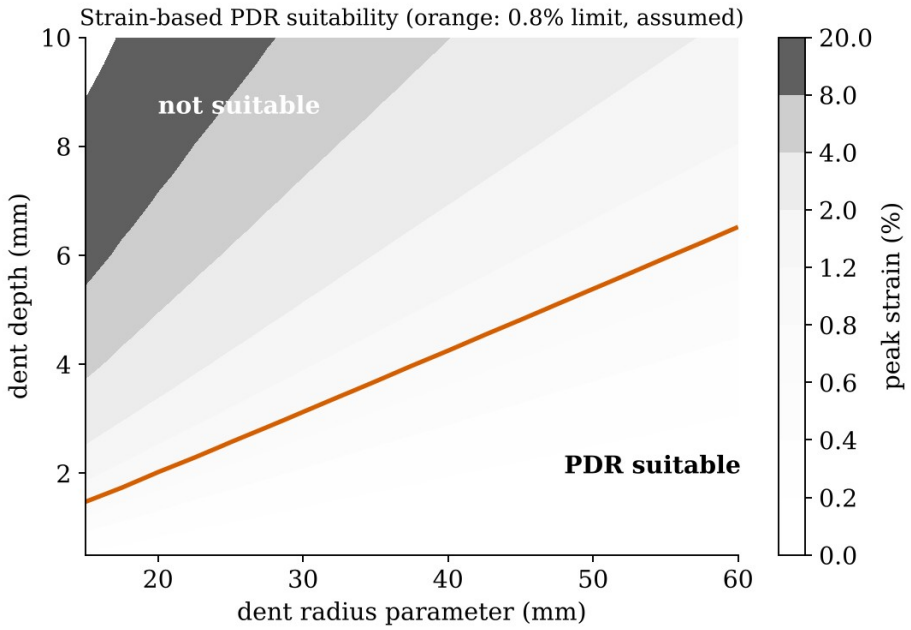


Figure 17: Peak strain map and the assumed PDR suitability limit

Table 20: End-to-end assessments produced by the prototype (simulated dents)

Case	Depth / size	Peak strain	P(success)	Recommendation and reason
Shallow door dent, rear access	1.5 mm / 110 mm	0.12%	1.00	Push repair: strain below limit; rear side accessible
Shallow roof dent, no rear access	1.8 mm / 124 mm	0.14%	1.00	Glue-pull: 20 mm tab, about 2 pulls, force 154 N
Sharp crease	6.0 mm / 43 mm	16.1%	0.00	Stud-pull + refinish: paint likely to crack
Dent with damaged paint	3.0 mm / 94 mm	0.56%	0.96	Stud-pull + refinish: paint already damaged

For 0.7 mm mild steel with a yield strength of 200 MPa the prototype gives a collapse force of 153.9 N and, with a safety factor of 2 and an adhesion strength of 1 MPa, a minimum glue-tab area of 308 mm² (about 20 mm diameter), consistent with common PDR tab sizes.

5.5 Prototype API and Tests

The FastAPI prototype exposes the full pipeline. A request to *POST /api/assess/synthetic* with a dent depth, size and panel description returns a JSON report containing depth, diameter, peak strain, suitability, success probability, method, reasons, force, tab size, number of pulls and a manual-check flag. The table below lists the automated tests.

Table 21: Automated tests (pytest 9.0, all passing)

ID	Test	What it verifies	Result
T1	test_phase_shift_recovers_phase	4-step algorithm returns the exact phase	Pass

ID	Test	What it verifies	Result
T2	test_reconstruction_depth_within_half_mm	Depth error ≤ 0.5 mm with noise and 2% calibration error	Pass
T3	test_plate_solver_matches_exact_solution	Solver within 3–4% of exact plate solutions	Pass
T4	test_flat_panel_is_suitable_and_deep_crease_is_not	Suitability check behaves correctly at both extremes	Pass
T5	test_yield_line_force_example	Pc = 153.9 N for 0.7 mm, 200 MPa	Pass
T6	test_method_selection_rules	Push, glue-pull, stud-pull and replacement chosen correctly	Pass
T7	test_conformal_coverage	Average interval coverage ≥ 0.895 for α = 0.1	Pass
T8	test_api_end_to_end	API health and full assessment request	Pass

5.6 Limitations of the Current Prototype

- All results are on simulated dents; real stripe capture on car paint has not yet been tested.
- The suitability limit and the glue and adhesion strengths are literature-based placeholders awaiting calibration.
- The plate model is linear-elastic; plastic behaviour is approximated through yield-line theory only.
- The single-photo network (C2) and the technician user interface are not yet implemented.

6. Project Plan

6.1 Work Breakdown Structure

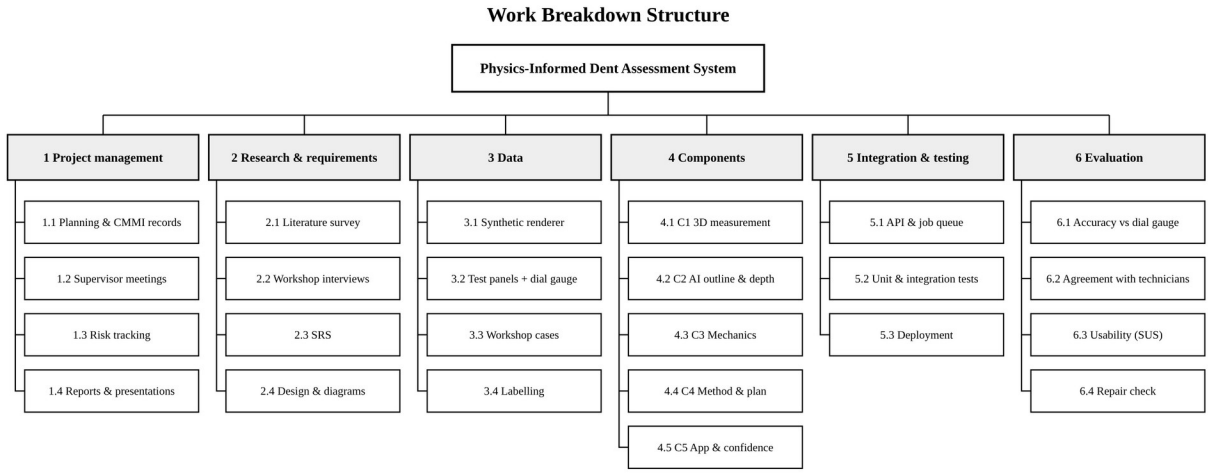


Figure 18: Work breakdown structure

6.2 Schedule and Milestones

The Gantt chart shows the schedule with progress as of 8 November 2026. Test panels are scheduled first because Components 1 and 3 need dial-gauge ground truth before they can be validated.

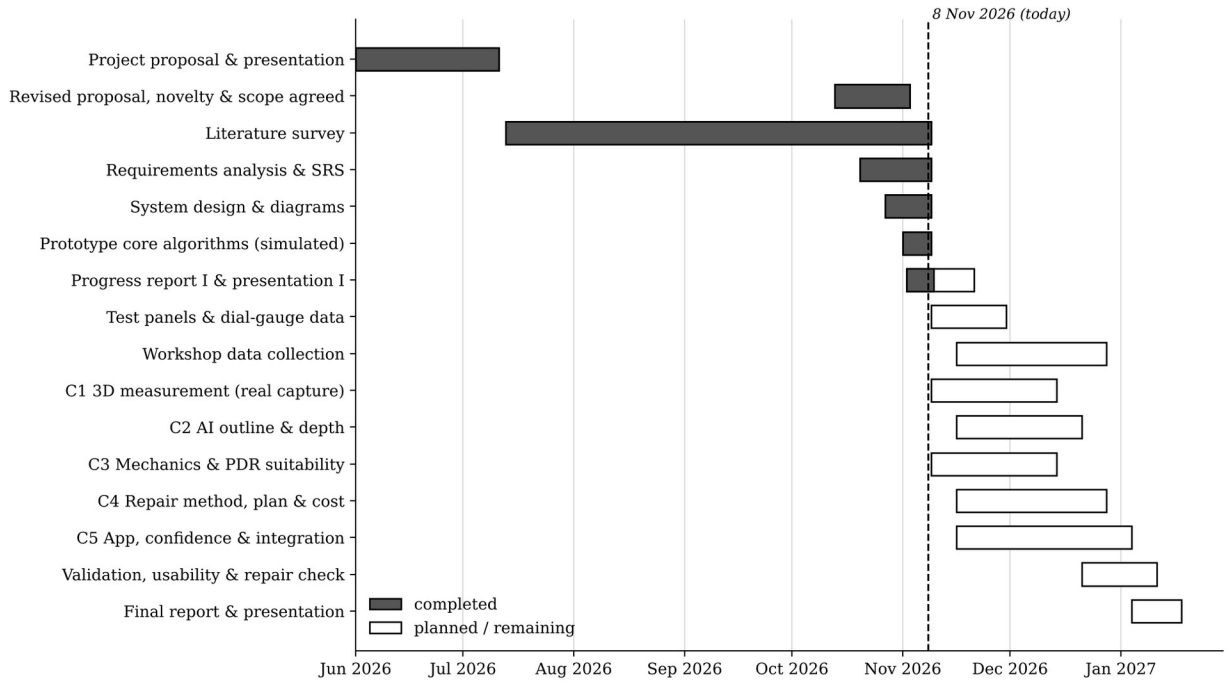


Figure 19: Project schedule (shaded: completed)

Table 22: Milestones

Milestone	Target date	Status
M1 Project proposal and presentation	Jul 2026	Achieved
M2 Revised proposal; novelty and scope agreed	02 Nov 2026	Achieved
M3 Progress report I (SRS, literature, design, plan)	08 Nov 2026	Achieved (this report)
M4 Progress presentation I	[date]	Planned
M5 Test-panel dataset and first real 3D scans	29 Nov 2026	Planned
M6 All components integrated	03 Jan 2027	Planned
M7 Evaluation complete (accuracy, usability, repair check)	10 Jan 2027	Planned
M8 Final report and presentation	17 Jan 2027	Planned

6.3 Work Distribution

Table 23: Work distribution

Member	Component / objective	Main algorithms and tasks
D.M.S.N. Bandara (Leader)	C1 3D measurement (O1); project management	Phase shifting, temporal unwrapping, slope calibration, Poisson integration; planning, CMMI records
S.D.K.S. David	C2 AI outline and depth (O2)	Synthetic renderer, U-Net mask and height network, synthetic-to-real fine-tuning
K.S. Dissanayaka	C3 Mechanics and PDR suitability (O3)	Curvature, strain maps, finite-difference plate solver, suitability check
U.S.N. Senevirathne	C4 Repair method, plan and cost (O4)	Yield-line force, glue-tab and pull calculations, push planning, plan-based costing
K.L.N.C. Liyanage	C5 App, confidence and integration (O5, O6)	Paired capture app, Monte Carlo, conformal intervals, API, deployment
All members	Validation (O7)	Test panels, dial-gauge measurement, workshop data, repair check, usability study, reports

6.4 Risk Register

Table 24: Risk register

Risk	Likelihood / impact	Mitigation	Owner
Poor stripe reflections (dirty, dull paint, sunlight)	Medium / High	Capture in shade, clean panel, quality checks, C2 fallback	Bandara
Phone movement between patterns	Medium / Medium	Tripod, frame alignment, automatic rejection	Bandara
Elastic model misses plastic behaviour	High / Medium	Yield-line corrections calibrated on test panels	Dissanayaka

Risk	Likelihood / impact	Mitigation	Owner
Paint cracking strain uncertain	High / Medium	Use published range; Monte Carlo reports it as probability	Dissanayaka
Large synthetic-to-real gap	Medium / Medium	Deflectometry labels real photos for fine-tuning	David
Too few workshop cases	Medium / High	Early workshop contact; test panels answer RQ1–RQ3	All
Equipment delay (dial gauge, panels)	Low / High	Order now; borrow from university lab	Senevirathne
Integration delays	Medium / Medium	Typed interfaces, continuous integration, weekly merges	Liyanage

7. CMMI Meeting Minutes

7.1 Process Approach

The team records its meetings following practices from CMMI for Development [35]. Each meeting is linked to the process areas it supports: **PP** Project Planning, **PMC** Project Monitoring and Control, **REQM** Requirements Management, **DAR** Decision Analysis and Resolution, and **RSKM** Risk Management. Every meeting records the agenda, discussion, decisions with their rationale, and action items with an owner and due date; open actions are reviewed at the start of the next meeting. Fields highlighted in yellow must be confirmed by the team before submission.

7.2 Meeting Log

Table 25: Meeting log

No.	Date	Type	Main outcome	CMMI areas
MM-01	[date]	Panel feedback review	Proposal to be revised: more depth and mathematical content	PMC, REQM
MM-02	[date]	Team meeting	New features defined; one component per member	PP, DAR
MM-03	[date]	Supervisor meeting	Repair (strength) check added: field + laboratory	REQM, DAR
MM-04	[date]	Supervisor meeting	Application justified; local prior art reviewed; novelty narrowed	REQM, DAR, RSKM
MM-05	[date]	Supervisor meeting	Novelty approved; workshops as users; dent scope agreed	REQM, PP
MM-06	[date]	Team meeting	Work distribution revised; Progress Report I tasks allocated	PP, PMC

7.3 Minutes of Meetings

Table 26: MM-01: Review of panel feedback on the initial proposal

Date / time	[date and time]
Type / venue	Panel / lecturer feedback session
Attendees	Group 03 members; [confirm attendance]
Agenda	1. Feedback received on the initial proposal. 2. Response strategy.
Discussion	The panel considered the initial scope (dent detection and cost estimation) too simple for five members and noted that it did not require significant AI or mathematical work. The team discussed adding measurement, mechanics and planning features that each member could own.
Decisions	Revise the proposal to increase depth and breadth with mathematically grounded components; prepare a revised proposal and presentation.
Action items	Draft revised feature list – all – [due]; Collect literature on 3D measurement and panel mechanics – all – [due]

CMMI process areas	PMC, REQM
Recorded by	[name]

Table 27: MM-02: Revised feature set and component allocation

Date / time	[date and time]
Type / venue	Team meeting
Attendees	All five members; [confirm]
Agenda	1. Candidate features. 2. Allocation to members. 3. Revised proposal documents.
Discussion	Features considered: 3D measurement by deflectometry, single-photo depth with AI, strain-based PDR suitability, repair-method selection and planning, confidence levels. Each was assessed for novelty, feasibility with low-cost equipment and testability with a dial gauge.
Decisions	Adopt five components (C1–C5), one per member, each with an algorithmic core; keep detection and cost estimation as supporting parts.
Action items	Revised proposal document – Bandara – [due]; Revised presentation – all – [due]
CMMI process areas	PP, DAR
Recorded by	[name]

Table 28: MM-03: Strength checking after repair; literature reading

Date / time	[date and time]
Type / venue	Supervisor meeting (online) [confirm mode]
Attendees	Supervisor; Group 03 members; [confirm]
Agenda	1. Supervisor question: where is the strength check after repair? 2. Papers read by the team.
Discussion	The team reported that workshops do not strength-test after PDR and that the IMI standard accepts a repair when no sign of the dent is visible. The supervisor asked why the project could not include such a check and whether it would add value. The team agreed that a check is useful if it does not damage customer vehicles.
Decisions	Add a repair check: non-destructive field check (3D re-scan, stiffness comparison with the opposite panel, AI paint-crack check) and laboratory strength tests on scrap test panels (re-dent, load–deflection, cross-cut adhesion).
Action items	Design repair-check procedure – Dissanayaka & Bandara – [due]; Each member to summarise key papers – all – [due]
CMMI process areas	REQM, DAR
Recorded by	[name]

Table 29: MM-04: Justification of the application and local prior art

Date / time	[date and time]
Type / venue	Supervisor meeting (online) [confirm mode]
Attendees	Supervisor; Group 03 members; [confirm]

Agenda	1. Issues caused by wrong dent assessment. 2. Existing Sri Lankan systems. 3. Novelty.
Discussion	The team explained the consequences of wrong assessment: cracked paint and rework, unnecessary repainting, inconsistent quotes and lack of evidence. A lecturer identified local prior art (Allianz Lanka 3D app, Click2Claim, LOLC V Claim, SLIC virtual claims, Fairfirst/CamCom AI, and a SLIIT 2025 paper). These detect damage or digitise claims but do not measure dents or decide PDR suitability.
Decisions	Narrow the novelty to 3D dent measurement, PDR suitability/repair-method decision and evidence-based decision support; add local prior art to the literature review. Risk noted: overlap with insurer AI tools.
Action items	Update literature review with local systems – David – [due]; Send novelty proposal to supervisor – Bandara – [due]
CMMI process areas	REQM, DAR, RSKM
Recorded by	[name]

Table 30: MM-05: Approval of novelty, users and scope

Date / time	[date and time]
Type / venue	Supervisor meeting (online) [confirm mode]
Attendees	Supervisor; Group 03 members; [confirm]
Agenda	1. Novelty focus. 2. Target users. 3. Meaning of “dent”.
Discussion	The supervisor agreed to focus on the proposed novelty. The team clarified that the system is designed for workshop technicians rather than insurers. The supervisor asked whether “dent” means all damage; the team proposed a scope limited to painted metal outer panels, up to about palm size, with or without paint damage.
Decisions	Novelty approved. Primary users: workshop technicians; reports shareable with insurers. Scope as in Section 1.3. Dent pulling (glue-pull, stud-pull) explicitly included.
Action items	Update proposal and presentation – Bandara – [due]; Concept paper – all – [due]
CMMI process areas	REQM, PP
Recorded by	[name]

Table 31: MM-06: Work distribution and Progress Report I

Date / time	[date and time]
Type / venue	Team meeting
Attendees	All five members; [confirm]
Agenda	1. Final component ownership. 2. Progress report requirements. 3. Implementation start.
Discussion	Ownership was rebalanced to match skills: 3D measurement to Bandara, AI depth to David, app and confidence to Liyanage. The progress report requirements (SRS, literature survey, diagrams, CMMI minutes, project plan, early implementation) were allocated.

Decisions	Adopt the work distribution in Section 6.3. Start implementation of core algorithms in Python with automated tests.
Action items	SRS – Liyanage & Senevirathne – [due]; Diagrams – all – [due]; Prototype algorithms – component owners – [due]; Order dial gauge and scrap panels – Senevirathne – [due]
CMMI process areas	PP, PMC
Recorded by	[name]

8. Conclusion and Next Steps

The project has moved from proposal to a specified, designed and partly implemented system. The literature survey confirms the gap: existing solutions, including those in Sri Lanka, detect damage in 2D and estimate cost, but none measure the dent, check repairability from mechanics or support the technician's choice between pushing and pulling. The SRS defines 32 functional and 18 non-functional requirements traced to the seven objectives, and the design is documented with the full set of UML, DFD, ER and deployment diagrams. The prototype shows that the core algorithms work on simulated data and that the plate solver is correctly implemented.

Next steps before Progress Presentation II:

1. Build the test-panel rig and measure at least 30 steel-ball dents with the dial gauge (all).
2. Capture real stripe sets on painted panels and compare with dial-gauge profiles (Bandara).
3. Implement the synthetic renderer and train the first U-Net (David).
4. Calibrate the strain limit on painted test panels (Dissanayaka).
5. Calibrate pull counts and tab sizes with real glue-pull trials (Senevirathne).
6. Build the technician capture interface and connect it to the API (Liyanage).
7. Begin workshop data collection with signed consent (all).

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